THE ADEQUACY FUNCTOR AS A COLIMIT OVER LINEARLY ADEQUATE FUNCTORS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Every adequate functor is a quotient of a direct sum of linearly adequate functors, and kernels of adequate maps are adequate. Same chapter, Lemmas lemma-adequate-surjection-from-linear and lemma-kernel-adequate, invoked at proof lines 1057-1064. Give the exact sequence K -> L -> G -> 0 used to reduce the universal property to linearly adequate test objects.
- Colimits of adequate functors are adequate. Same chapter, Lemma lemma-colimit-adequate, invoked at proof lines 1080-1082. Makes Q(F) adequate.

Source reference. Original lemma 06UX: sources/stacks-adequate/pinned-original-source.tex, source lines 468–468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V4: sources/stacks-adequate/pinned-original-source.tex, source lines 734–734. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V3: sources/stacks-adequate/pinned-original-source.tex, source lines 709–709. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. For every module-valued functor F on Alg_A there exists a morphism $Q(F) \to F$ of module-valued functors on Alg_A such that (1) $Q(F)$ is adequate and (2) for every adequate functor G the map* $\mathrm{Hom}(G, Q(F)) \to \mathrm{Hom}(G, F)$ *is a bijection.*

Proof. Choose a set $\{L_i\}_{i \in I}$ of linearly adequate functors such that every linearly adequate functor is isomorphic to one of the L_i. This is possible. Suppose that we can find $Q(F) \to F$ with (1) and (2)' or every $i \in I$ the map $\mathrm{Hom}(L_i, Q(F)) \to$

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$\mathrm{Hom}(L_i, F)$ is a bijection. Then (2) holds. Namely, combining Lemmas 06UX and 06V3 we see that every adequate functor G sits in an exact sequence

$$K \to L \to G \to 0$$

with K and L direct sums of linearly adequate functors. Hence (2)' implies that $\mathrm{Hom}(L, Q(F)) \to \mathrm{Hom}(L, F)$ and $\mathrm{Hom}(K, Q(F)) \to \mathrm{Hom}(K, F)$ are bijections, whence the same thing for G.

Consider the category $\mathcal{I}$ whose objects are pairs (i, φ) where $i \in I$ and $\varphi : L_i \to F$ is a morphism. A morphism $(i, \varphi) \to (i', \varphi')$ is a map $\psi : L_i \to L_{i'}$ such that $\varphi' \circ \psi = \varphi$. Set

$$Q(F) = \mathrm{colim}_{(i,\varphi) \in \mathrm{Ob}(\mathcal{I})} L_i$$

There is a natural map $Q(F) \to F$, by Lemma 06V4 it is adequate, and by construction it has property (2)'. $\square$